

Chapter 1: An Introduction to Chemistry

What is Chemistry?

Chemistry - The study of the structure and behavior of matter

Matter – Anything that occupies space.

- Can be a solid, liquid, or gas

The Scientific Method

Recognize the problem and state it clearly

- Make an **observation**

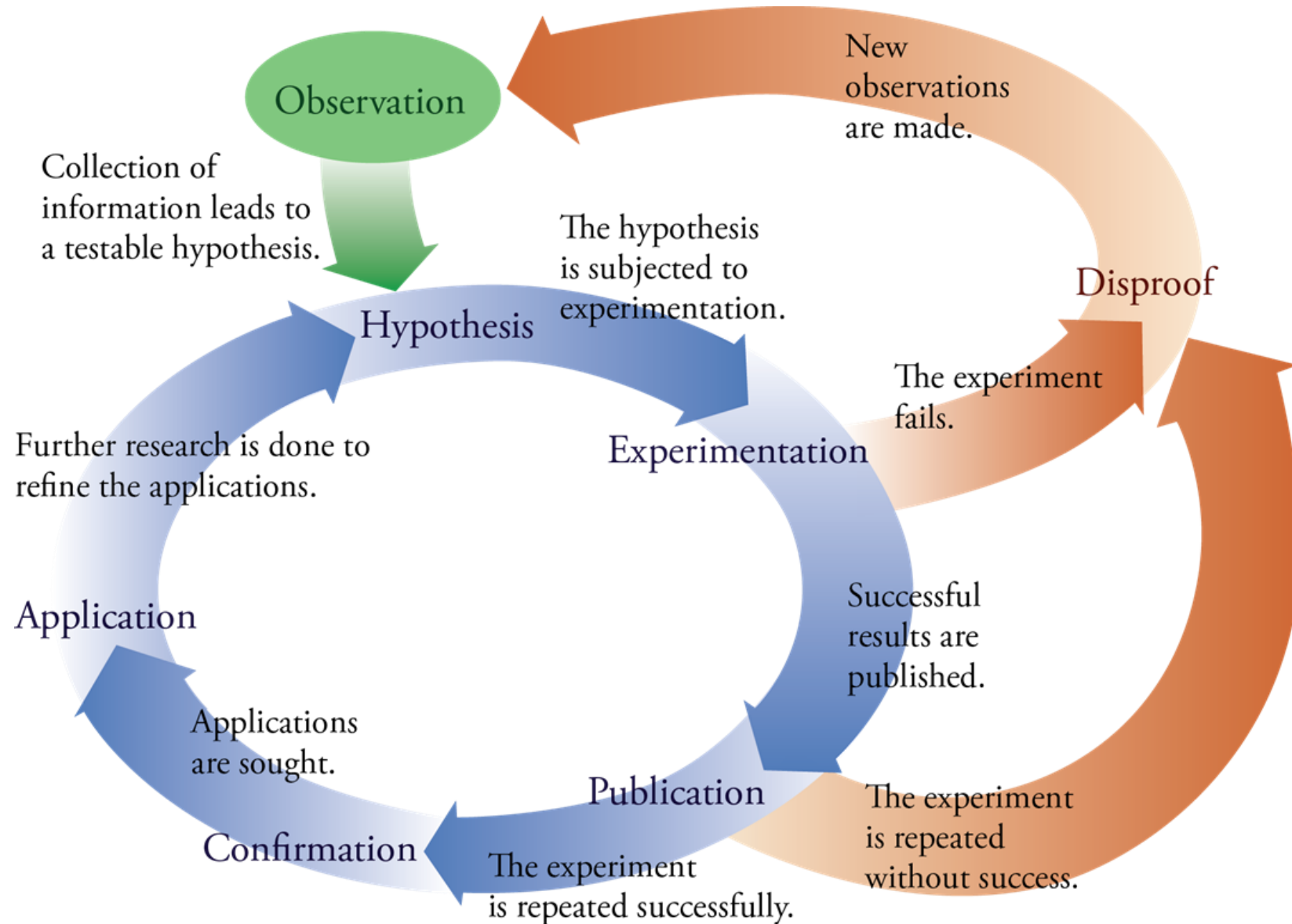
Propose possible solutions to the problem or possible explanations for the observation

- Formulate a **hypothesis**

Decide which of the solutions is the best, or decide whether the explanation proposed is reasonable

- Perform an **experiment**

The Scientific Method



The Scientific Method

Theory is a well tested hypothesis

Law is a statement of a natural phenomenon without explanation of the process

Measurements and Units

Value – a quantitative description that includes a number and a unit

- How a measurement is reported

Units – quantities defined by a set of agreed upon standards that can compare an event or object to another.

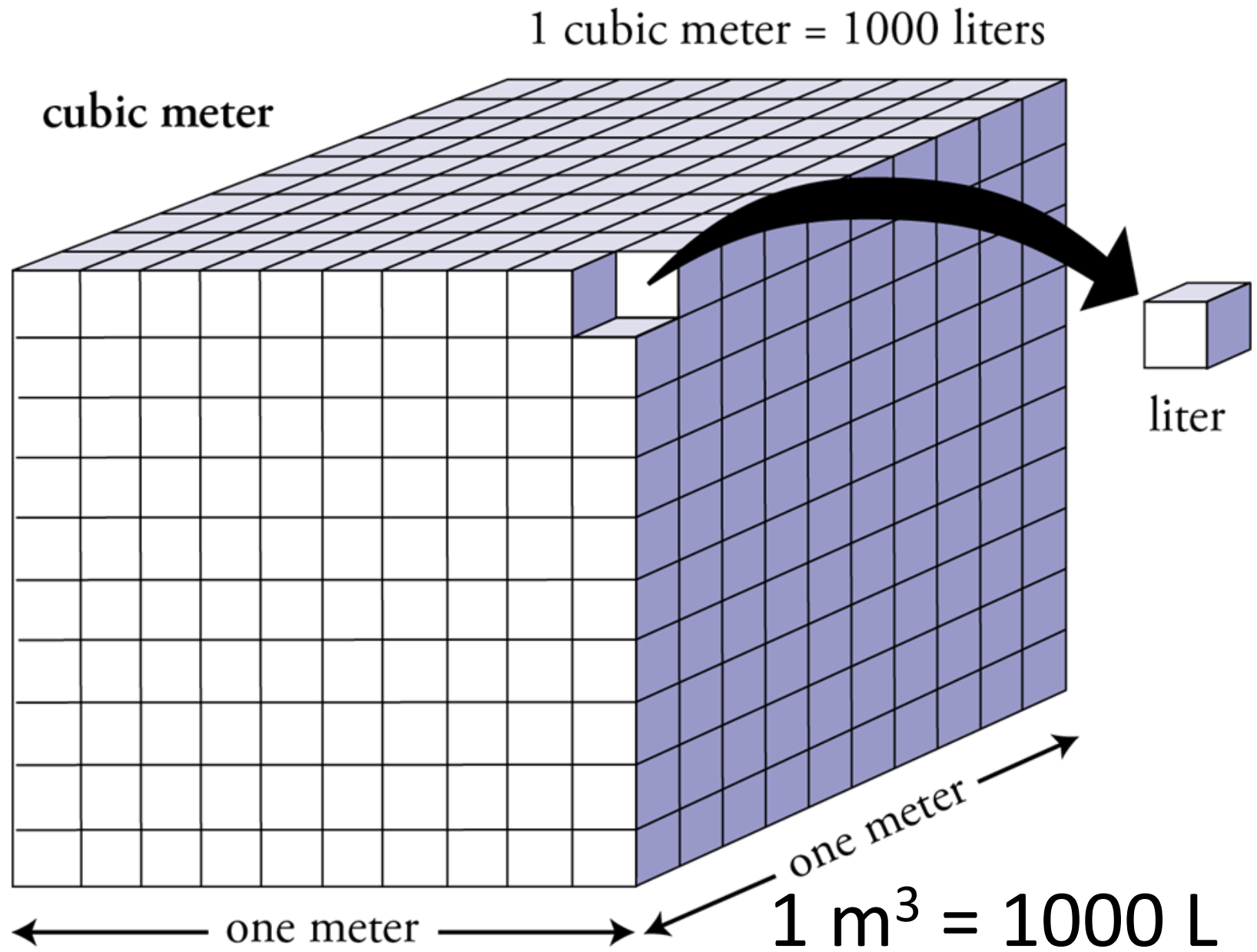
The International System of Measurement (SI System)

Measurement Type	Base Unit	Abbreviation
Length	meter	m
Mass	kilogram	kg
Time	second	s
Temperature	Kelvin	K
Amount of a substance	mole	mol
Electric current	ampere	A
Luminous intensity	candela	cd

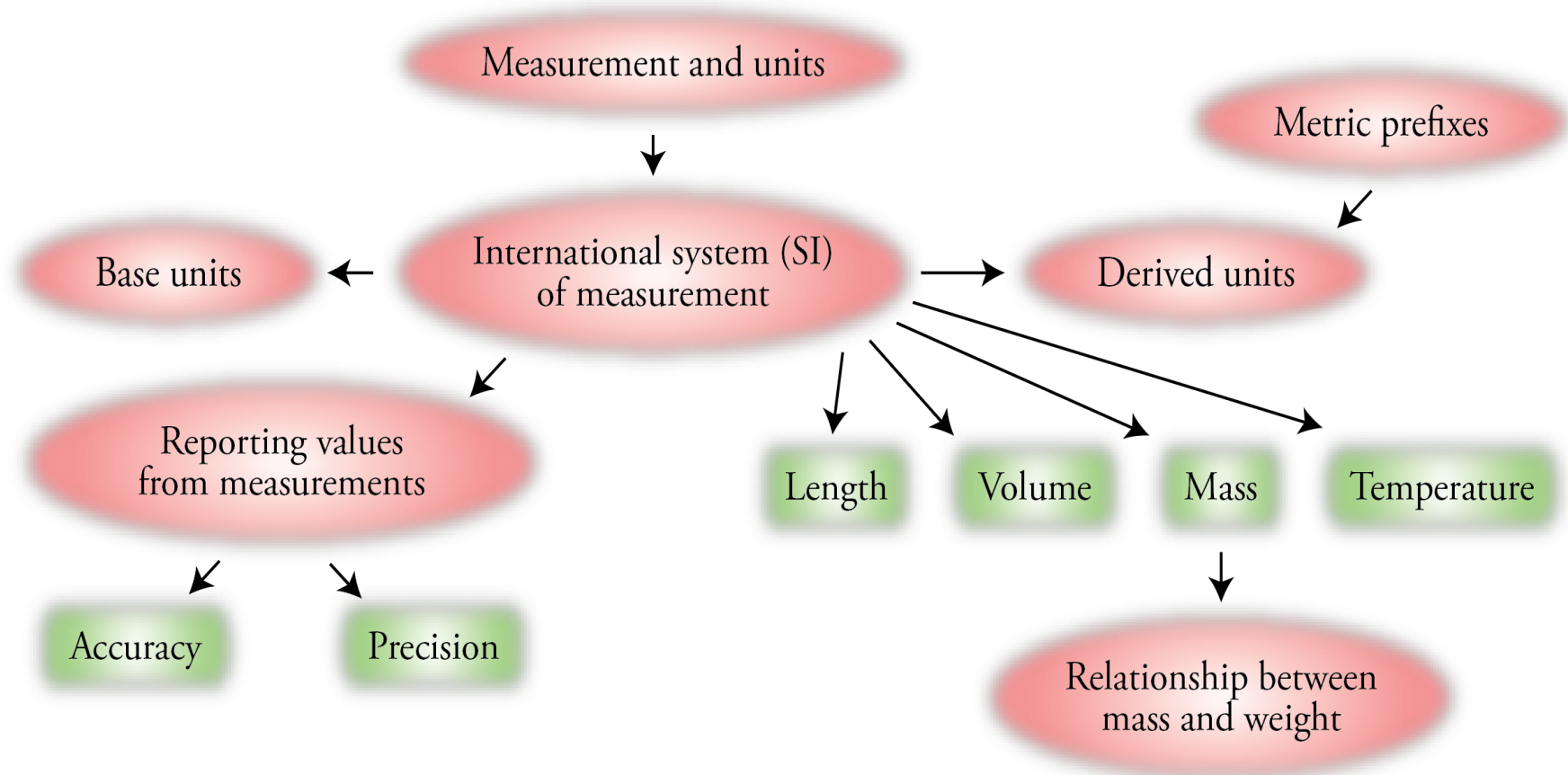
Metric Prefixes

Prefix	Abbreviation	Number
giga	G	10^9 or 1,000,000,000
mega	M	10^6 or 1,000,000
kilo	k	10^3 or 1000
centi	c	10^{-2} or 0.01
milli	m	10^{-3} or 0.001
micro	μ	10^{-6} or 0.000001
nano	n	10^{-9} or 0.000000001
pico	p	10^{-12} or 0.000000000001

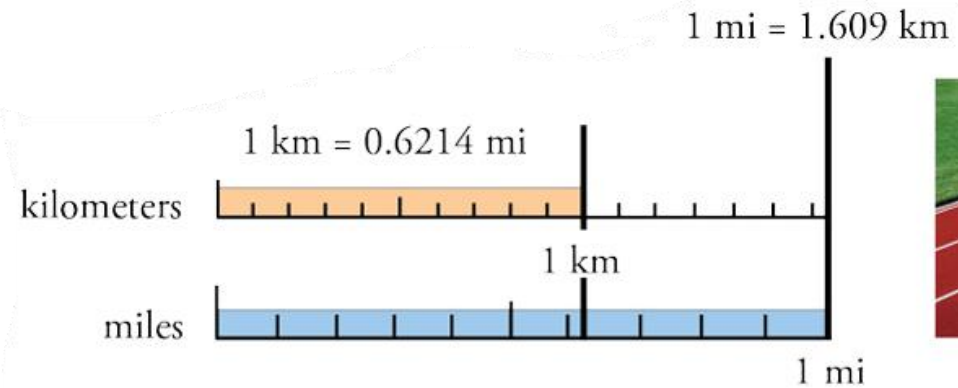
SI Units Derived from Base Units



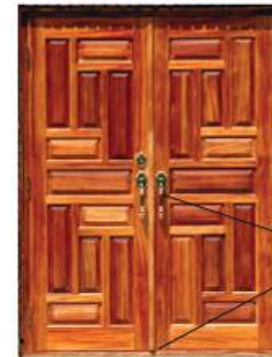
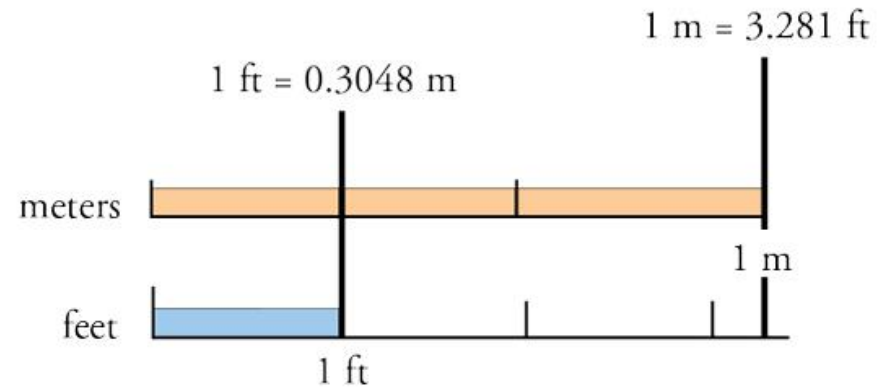
Unit Map



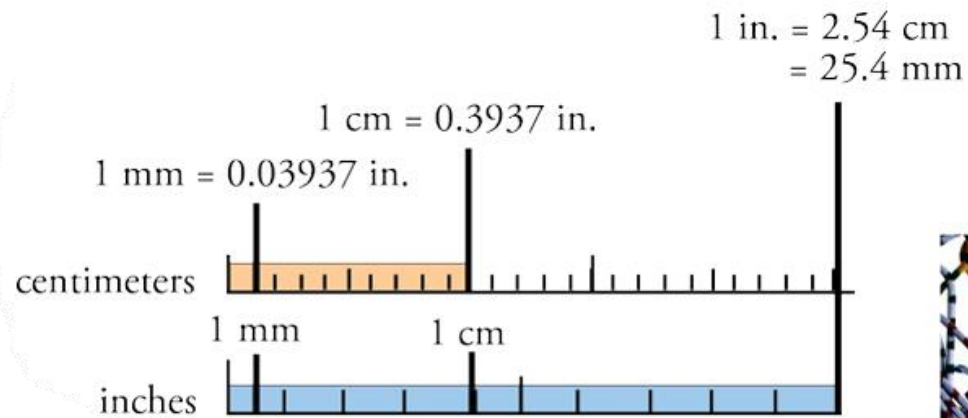
Relationships Between Units of Length



A mile is four times around a typical high school track.



1 meter



1 centimeter



1 millimeter

Relationships Between Units of Volume

$$1 \text{ cm}^3 = 1 \text{ mL}$$

1 fluid ounce (fl oz)



1 fl oz = 29.57 mL

1 mL = 0.03381 fl oz



1 milliliter (mL)
= about 20 drops

1 gal = 3.785 L



1 gallon (gal)
or 4 quarts (qt)

1 qt = 0.9464 L



1 qt or 32 fl oz

1 L = 1.057 qt
= 0.2642 gal



1 liter (L)
or 1000 mL

Mass and Weight

Mass – measurement of the amount of matter in an object.

- Property of matter that leads to gravitational attraction between objects

Weight – measurement of the force of gravitational attraction of an object and a significantly larger body.

Relationships Between Units of Mass

$$1 \text{ lb} = 453.6 \text{ g}$$

$$1 \text{ kg} = 2.205 \text{ lb}$$

$$1 \text{ oz} = 28.35 \text{ g}$$



About 2.5 grams (g) or
about 0.088 ounce (oz)



About 1 kilogram (kg) or
about 2.2 pounds (lb)

$$1 \text{ Mg} = 1000 \text{ kg} = 1 \text{ t}$$



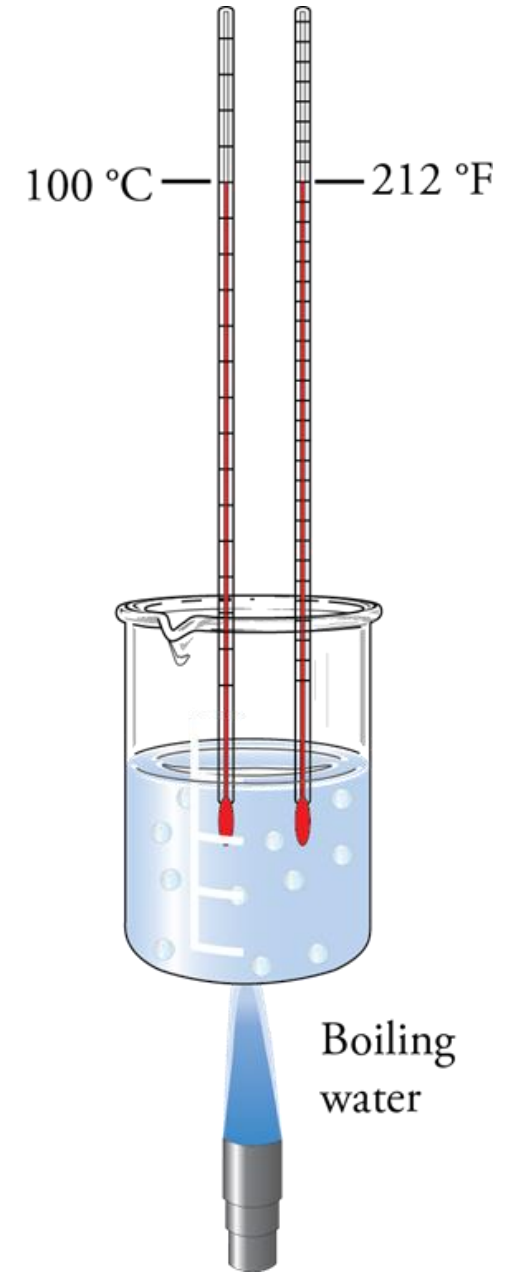
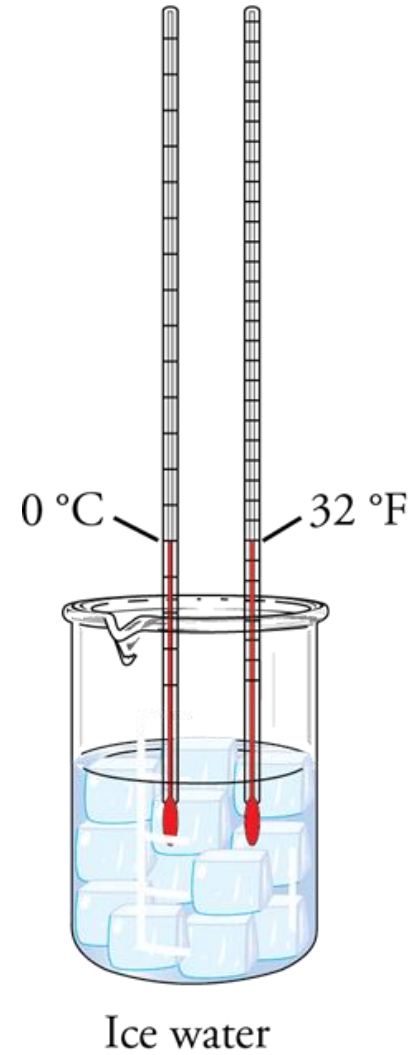
About 1 megagram (Mg) or 1 metric ton (t)

Temperature

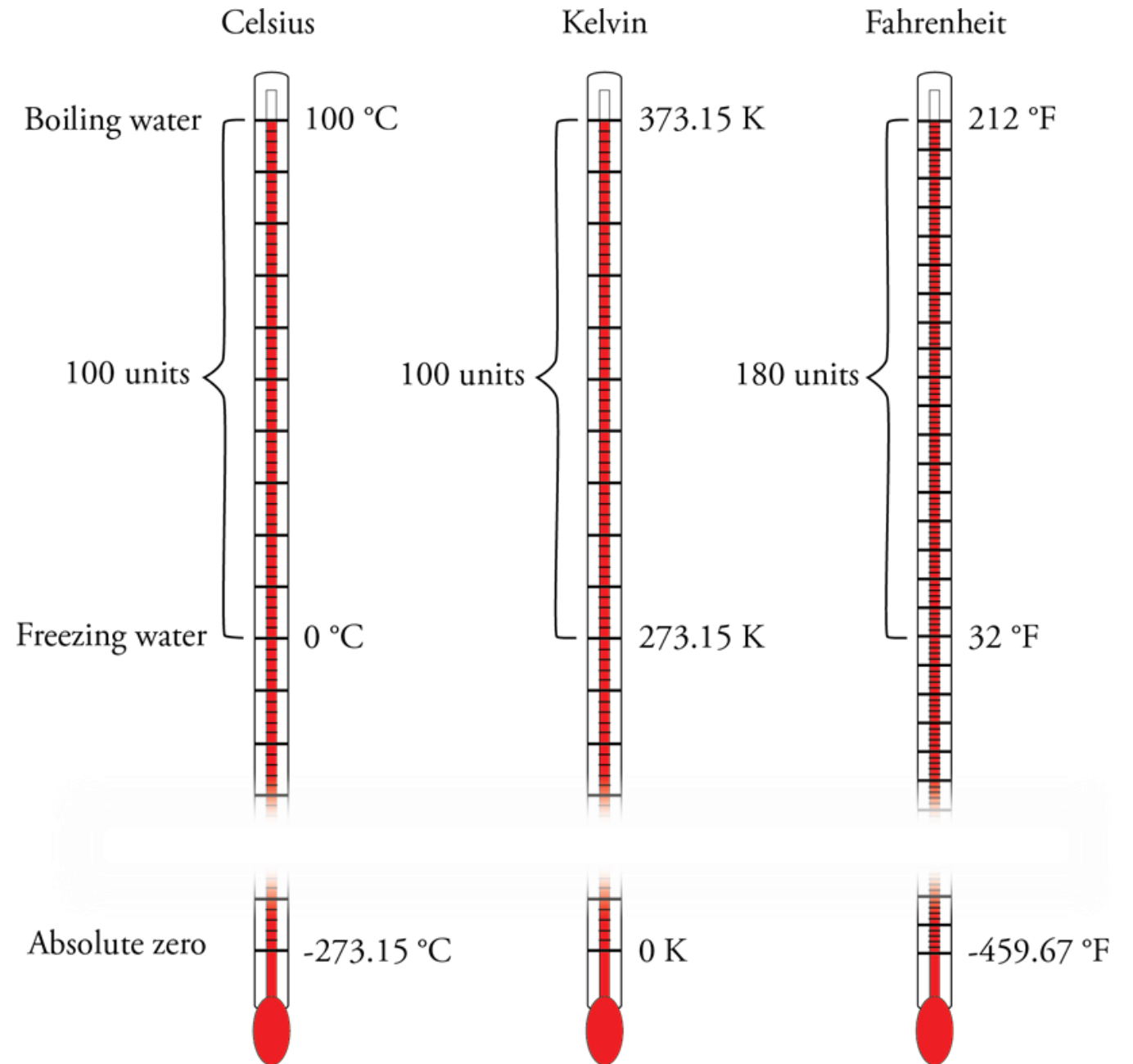
Measures the average motion of particles within a system

- The **higher** the temperature the **faster** the particles are moving
- The **lower** the temperature the **slower** the particles are moving

Temperature

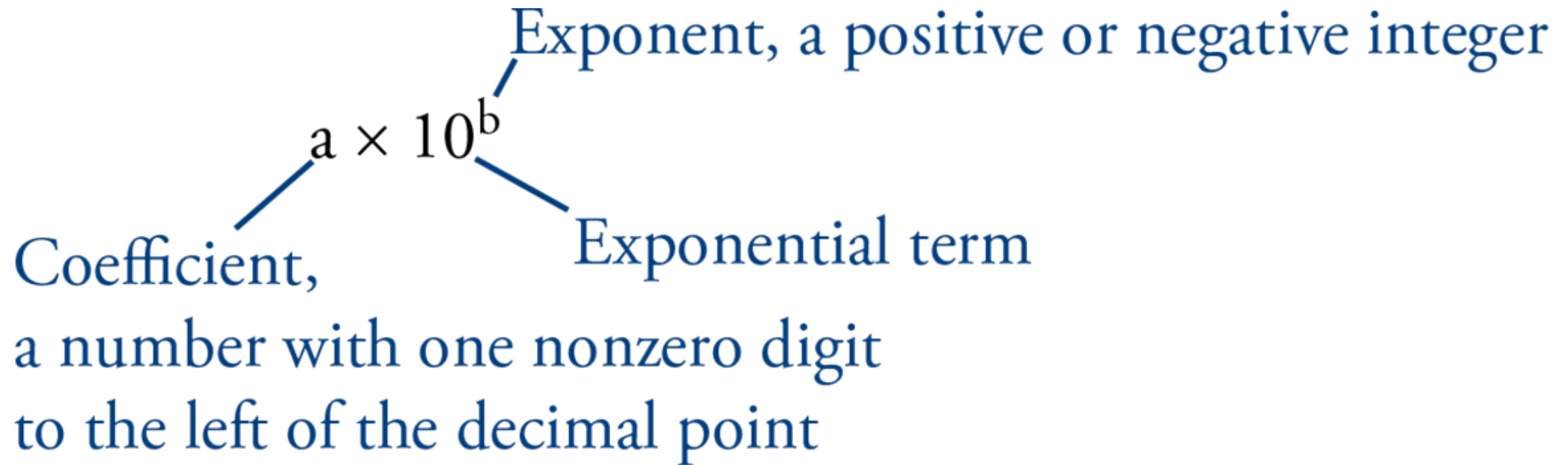


Comparing Temperature Scales



Scientific Notation

When expressing numbers in scientific notation the following conventions are used:



Scientific Notation – Large Numbers

When expressing large numbers in scientific notation the decimal moves to the **left**

10500000000000.



Scientific Notation – Small Numbers

When expressing small numbers in scientific notation the decimal moves to the **right**

0.0000000004060

